

CH:4 - Random Numbers

④ A random number that is drawn from a uniformly distributed random variable between some specified intervals, and they have equal probability of occurrence.

④ ⇒ Properties of random numbers:

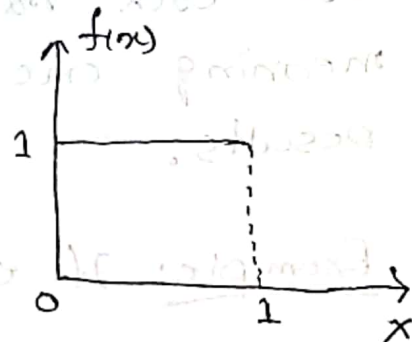
A sequence of random numbers has two important statistical properties.

- Uniformity
- Independence

Each random number is an independent sample drawn from a continuous uniform distribution between an interval 0 to 1.

The probability density function

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



The expected value of each random number

R_i is given by

$$E(R) = \int_0^1 f(x) dx = \left| \frac{x^2}{2} \right|_0^1 = \frac{1}{2}$$

and variance

$$\begin{aligned} V(R) &= \int_0^1 x^2 dx - [E(R)]^2 = \left| \frac{x^3}{3} \right|_0^1 - \left[\frac{1}{2} \right]^2 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

What are the purpose of testing uniformity and independence of random values?

Soln: The purpose of testing uniformity is to check if random values are spread evenly across a range, meaning every outcome is equally likely.

Example: Roll a die 60 times. If each number (1-6) shows up around 10 times, it's uniform.

The purpose of testing independence ensures that each random value doesn't affect the others, meaning one result doesn't depend on previous results.

Example: If each roll doesn't affect the next (e.g., a 4 can follow a 1), it's independent.

Why testing uniformity and independence are necessary →

1. Ensure values are evenly distributed.
2. Prevent biases that could affect security.
3. Value is not dependent on previous values.
4. Prevents patterns that could lead to predictability.

④ Pseudo Random Numbers:

A random number that are generated by using some known arithmetic operation.

Since the arithmetic operation is known and the sequence of random numbers can be repeatedly obtained, the numbers cannot be called truly random.

Disadvantages:

- ① May not be uniformly distributed.
- * May not be continuous.
- * The mean of the generated numbers may be too high or too low.
- * The variance may be too high or too low.
- * There may be cyclic patterns, like;
 - ② Auto correlation between numbers.
 - ③ A group of numbers continuously above the mean.

7 $\Rightarrow$ Method: (a)

* Congruence or Residue

3 type $\rightarrow$ additive $\Rightarrow r_{i+1} = (r_i + b) \text{ modulo } M$

multiplication $\Rightarrow r_{i+1} = ar_i \text{ modulo } M$

recursive $\Rightarrow r_{i+1} = (ar_i + b) \text{ modulo } M$

- Disadvantage of congruence method $\rightarrow$

1. Not possible to generate a non repeating sequence.

2. Number obtained are not truly random because the number can be predicted rather computed from r_i and whole string is reproducible.

Q. Why the random numbers generated by computer are called ~~pseudo~~ pseudo random number? Discuss the pitfalls of congruence method of generating random numbers.

Sol: Computer-generated random numbers are called pseudo-random because they aren't truly random but are produced by algorithms that mimic randomness. They follow a predictable pattern, but for most practical purposes, they appear random.

D.M

The congruence method for generating random numbers has some pitfalls:-

1. Patterns:

It can produce numbers in repeating patterns which can reduce randomness.

2. Limited Range:

The numbers might be limited to a small range, making them less useful.

3. Predictable: If you know the starting point (seed), you can predict future numbers.

For example, if you use the same seed, you might get the same sequence of numbers every time, which isn't truly random.

9.51 Mid-Square Random Number Generator:

↳ The method comprises of starting with an n -digit number, squaring it and taking the n digits in the middle as the next number,



⊙ Qualities of an efficient random number generator.

- It should generate a sufficiently long sequence of random number.
- The random numbers generated by should be replicable.
- The generated random numbers should fulfill the requirements of uniformity and independence.
- The random number generator should be fast and cost-effective.
- It should be portable to different computers and ideally to different programming language.

The Kolmogorov-Simionov Test:

This test compares the continuous cdf, $F(x)$ of the uniform distribution to the empirical cdf, $S_n(x)$, of the sample of N random numbers.

Ex: 4.2 Hence, level of significance $\alpha = 0.05$

random number $\rightarrow .29, .89, .11, .61, .23, .86, .91,$

$.69, .50, .65$

$N = 10$

The calculation of the test,

Top row R_i random number in ascending order

2nd row compute empirical distribution, i/N

3rd row deviation, $(i/N - R_i)$ is computed. maximum of which gives D^+

4th row the $R_i - \frac{(i-1)}{N}$ is computed the maximum of which given D^- .

The largest deviation, $D = \max(D^+, D^-)$

R_i	.11	.23	.29	.41	.50	.61	.69	.65	.86	.89
i/N	.10	.20	.30	.40	.50	.60	.70	.80	.90	1.0
$D^+ \quad i/N - R_i$	-	-	.06	-	.00	-	.06	.15	.09	.11
$D^- \quad R_i - \frac{i-1}{N}$	.11	.13	.09	.11	.40	.11	.09	-	.06	-

Hence, $D^+ = 0.15$ $D^- = 0.13$

$D = 0.15$

7 The critical value of D for $\alpha=0.05$ and $N=10$ is 0.416 and the computed value is 0.15 is less than the critical value, the given random numbers are uniform at 95% level of significant.

* Chi-Square Test:

Chi-square is a characteristic of the distribution, which is a measure of its randomness.

$$\chi^2 = \sum_{i=1}^n \frac{(O - E)^2}{E}$$

Ex:

Class	frequency (O)	(O - E)	(Diff) ²
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$$\text{chi-square} = \frac{\sum (O - E)^2}{N}$$

Degree of freedom = (Number of class - 1)
Critical value at $\alpha=0.05$ is 16.919

At 95% confidence level the acceptable value of chi-square for 9 degree of freedom is 16.919 and for 99% confidence level — is 21.666

which is more than (chi-square).

Hence the random numbers are uniformly distributed.

$$\text{Confidence level} = 1 - \alpha$$

H Define Autocorrelation:

The uniformity test of random numbers is only a necessary test for randomness, not a sufficient one.

A sequence of number may be perfectly uniform and still not random. For example, the 1 2 3 4 5 6 7 8 9 12 3 would give a perfectly uniform distribution with the chi-square value is zero. But the sequence can by no means be regarded as random.

- ④ These given 100 random number given 99 pairs. Random number are divided into smaller number of classes as 3. Three class will be: less than equal 0.33, less than or equal 0.67 and less than or equal 1.0. With 3 classes in row and 3 class in columns, there will be total 9 cells. So 99 pairs, are grouped into 9 classes with expectation of 11 in each group.

P.T.O

Class	Frequency	Diff	(Diff) ²
$R_1 \leq .33 \ \& \ R_2 \leq .33$	17	6	36
$R_1 \leq .67 \ \& \ R_2 \leq .33$	6	5	25
$R_1 \leq 1.0 \ \& \ R_2 \leq .33$	13	2	4
$R_1 \leq .33 \ \& \ R_2 \leq .67$	7	4	16
$R_1 \leq .67 \ \& \ R_2 \leq .67$	12	1	1
$R_1 \leq 1.0 \ \& \ R_2 \leq .67$	9	2	4
$R_1 \leq .33 \ \& \ R_2 \leq 1.0$	15	4	16
$R_1 \leq .67 \ \& \ R_2 \leq 1.0$	10	1	1
$R_1 \leq 1.0 \ \& \ R_2 \leq 1.0$	15	0	0
	100		103

$$\text{Chi-Square} = \frac{103}{10} = 10.3$$

The criterion value of χ^2 for 8 degree of freedom at 99% confidence level is 20.09 which is less than the value we have obtained for the given sequence. Thus the given sequence of random number is serially autocorrelated at 99% confidence level.

4 Poker Test :

This test not only tests the randomness of the sequence of numbers, but also the digits comprising of each numbers.

Ex: 4.5

need to find logic -

five different digit 3029

Pairs 5040

Two pairs 1080

Three of a kind 720

full houses 90

four of kinds 45

five of a kind 1

The calculation χ^2 are : -

combination distribution	observed Distribution (O_i)	Expected E_i	$\frac{(O_i - E_i)^2}{E_i}$
five different digits	3075	3029	0.8601
Pairs	4935	5040	2.1875
Two pairs	1135	1080	2.8750
Three of kind	695	720	0.868
Full houses	105	90	2.5
Four of kind	59	45	1.8
five of kinds	1	1	0.0
	10,000	10,000	10.0907

7 The appropriate degree of freedom 6.

The critical value of χ^2 for 6 at $\alpha = 0.01$ is 16.8.

The value of χ^2 obtained from calculation is 10.097.

The hypothesis of independence cannot be rejected on the basis of p-value test.

Ex: 9.16 for four digit

$$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{1 \times 2 \times 3 \times 2}{2 \times 1 \times 2 \times 1} = 6$$

a) Four different digit probability is

$$P(a) = 0.9 \times 0.8 \times 0.7 \times 0.6 = 0.504$$

b) One Pair in the digits probability is

$$P(b) = \binom{4}{2} \times 0.1 \times 0.9 \times 0.8 \times 0.7 = 0.432$$

c) Two pair probability is

$$P(c) = \binom{4}{2} \times 0.1 \times 0.1 \times 0.9 \times 0.8 = 0.054$$

d) Three digits of one kind is

$$P(d) = 0.1 \times 0.1 \times 0.1 \times 0.9 = 0.009$$

e) Four digits of one kind is

$$P(e) = 0.1 \times 0.1 \times 0.1 \times 0.1 = 0.001$$

Combination Distribution	Observed Dis. O_i	Expected D_i, E_i	$\frac{(O_i - E_i)^2}{E_i}$
Four digits diff	5120	5040	1.2698
Pairs	4230	4320	1.8750
Two pairs	560	540	0.7407
Three of a kind	75	90	2.5
Four of a kind	15	10	2.5
	10,000	10,000	8.8856

Degree of freedom is 4

The critical value of $\chi^2_{0.05, 4} = 9.49$

So cannot be rejected on the basis of p-value test.

Ex: 4.11 In case of three-digit number —

- (a) All three of different kind
- (b) One pair
- (c) All three of same kind

$$P(a) = 0.9 \times 0.8 \times 0.7 = 0.504$$

$$P(b) = \binom{3}{2} \times 0.1 \times 0.9 \times 0.8 = 0.216$$

$$P(c) = 0.1 \times 0.1 \times 0.1 = 0.001$$

O_i	f_i	$\frac{(O_i - f_i)^2}{f_i}$
680	720	2.2222
289	270	1.3370
31	10	44.10
1000	1000	97.6592

degree of freedom = 2

$$\chi^2_{0.05, 2} = 5.99 < 97.6592$$

numbers is rejected.